CS450

Structure of Higher Level Languages

Lecture 27: Conclusion

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Thank you!

Novelties in CS450

- Introduction of **pattern matching** in Module 2
  - less boilerplate
  - no need for getters/setters
- Introduction of **TypedRacket** in Module 3
  - fewer students in office hours with type mismatch issues
  - some students with type mismatch errors (easier to handle, harder to discuss the output of a test case)
  - types + monad introduce friction (poor inference algorithm), but ultimately for good cause
- Modules were redesigned to have exactly 3 lectures (trying to make workload comparable across modules)
- **New module** to emphasize recursion (give more time to absorb recursion)

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Novelties in CS450

What was developed

- Refreshed/rewrote homework assignments HW2-9 (due to types, patten matching, and course restructuring)
- New slides/homework for module on optimized recursion (due to module restructuring; new module)
- Rewrite of slides Lecture 4-27 (due to types, pattern matching, course restructuring)
- A new homework assignment to emphasize recursion and fold (give more time to practice recursion)
- New infrastructure for test cases (group test-cases per exercise; allow turning off exercises)
- More test cases in HW7

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Novelties in CS450

Redesigned exercises

- Improve the focus on homework assignments
- *Approach:* more exercises that introduce a new learning concept on an already known algorithm
  - Optimized recursion revisit simple recursion (new)
  - Delayed evaluation revisit optimized recursion (new)
  - Monads revisit recursion (new)
  - Iterative programming language implementation: (i) expressions, (ii) function calls, (iii) out-of-order definitions
  - JavaScript's object module reuses the concepts of Racket's definitions

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